

Power Up **ANSWER KEY**

Early finishers · Grade 7 Accelerated · from Unit 2 into Unit 3

Name _____

1. Words for tomorrow

In 2^5 the 2 is the **base** and the 5 is the **exponent**.

2. Squares and cubes (from Unit 2)

n	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
n^2	1	4	9	16	25	36	49	64	81	100	121	144	169	196	225

n	1	2	3	4	5
n^3	1	8	27	64	125

3. Powers of 2 (each one is double the one before)

n	1	2	3	4	5	6	7	8	9	10
2^n	2	4	8	16	32	64	128	256	512	1024

4. Pattern hunt (use your powers-of-2 table)

- a) $2^2 \times 2^3 = 4 \times 8 = 32 = 2^5$
b) $2^3 \times 2^4 = 8 \times 16 = 128 = 2^7$
c) $2^1 \times 2^8 = 2 \times 256 = 512 = 2^9$
d) $2^5 \times 2^5 = 32 \times 32 = 1024 = 2^{10}$

What do you notice about the exponents? **When you multiply powers of 2, you add the exponents.**

5. Which is larger? Circle the larger one, or write = if they are equal.

- a) 2^5 or 5^2 b) 3^4 or 4^3 c) 2^4 or 4^2 d) 2^{10} or 10^3 e) 1^9 or 9^1
 2^5 (32 vs 25) 3^4 (81 vs 64) equal: both 16 2^{10} (1024 vs 1000) 9^1 (9 vs 1)

Challenge

Find a power between 500 and 600, using an exponent greater than 1. There are four of them. Can you find them all?

$2^9 = 512$, $8^3 = 512$, $23^2 = 529$, $24^2 = 576$

For you: $2^9 = 8^3$ because $8 = 2^3$, so $8^3 = (2^3)^3 = 2^9$: a preview of tomorrow's power of a power law. Part 4 previews the product of powers law.